
MVG-CRPS: A Robust Loss Function for Multivariate Probabilistic Forecasting

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Abstract

Multivariate probabilistic forecasting typically leverages neural network-based distributional regression, often employing Gaussian assumptions to simplify computation. While the standard negative log-likelihood provides analytical convenience, its sensitivity to outliers can severely degrade forecasting accuracy. Conversely, robust alternatives like the Energy Score, although less sensitive to extreme values, rely heavily on computationally expensive sampling approximations, limiting scalability in neural network training. To bridge this gap, we introduce the MVG-CRPS, a novel, strictly proper scoring rule for multivariate Gaussian distributions that maintains robustness to outliers while providing a closed-form expression, enabling efficient training and evaluation. Our approach leverages a whitening transformation, decorrelating multivariate outputs and reducing the multivariate scoring task to tractable univariate CRPS computations. Experiments on real-world datasets for both multivariate autoregressive and univariate sequence-to-sequence (Seq2Seq) forecasting tasks demonstrate that MVG-CRPS enhances robustness and predictive performance.

1 Introduction

Probabilistic forecasting is critical in applications ranging from financial risk management [1], to weather forecasting [2] and healthcare analytics [3], where accurate quantification of predictive uncertainty directly informs decision-making. Multivariate probabilistic forecasting models extend beyond point estimates, producing joint probability distributions across multiple correlated continuous variables. Neural network-based methods have become a dominant paradigm due to their flexibility and expressiveness [4–6]. Typically, these methods rely on parametric assumptions such as multivariate Gaussian distributions, allowing closed-form loss computations (e.g., log-likelihood) and efficient backpropagation.

Despite widespread adoption, standard metrics for model inference such as the negative log-likelihood (log-score) present substantial challenges. Most notably, under the Gaussian family, the log-score heavily penalizes unlikely events and outliers due to its exponential sensitivity in the tails of distributions, making it excessively sensitive to anomalies and model misspecification [7, 8]. As a result, neural network models trained using the log-score can generate overly conservative or inaccurate predictive distributions when exposed to real-world data characterized by occasional extreme events.

To address the limitations of the log-score, the Energy Score [ES, 9] emerged as a popular robust alternative. It generalizes the continuous ranked probability score [CRPS, 10, 11] for univariate distributions and effectively mitigates sensitivity to outliers by evaluating forecasts through expected pairwise distances between predictions and observations. However, the ES lacks a closed-form analytical expression in most cases, necessitating computationally intensive Monte Carlo sampling to

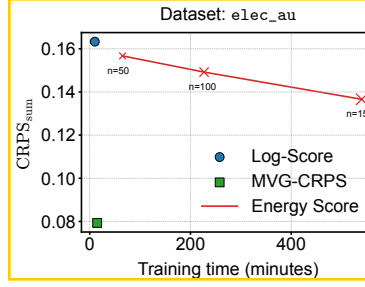


Figure 1: An example showing MVG-CRPS achieves better accuracy and faster training by avoiding sampling and reducing sensitivity to outliers. ES results are shown for different sample sizes.

approximate its value and gradients. Such approximations significantly slow down neural network training, limiting practical scalability [12, 13].

Motivated by the need for a robust yet computationally efficient scoring rule, this paper introduces MVG-CRPS (Multivariate Gaussian CRPS). We propose a strictly proper scoring rule specifically designed for multivariate Gaussian probabilistic forecasting tasks. Our approach circumvents the computational limitations of the ES by leveraging a PCA whitening transformation, decomposing the multivariate Gaussian distribution into independent, standard normal variables. Consequently, the multivariate scoring problem reduces to a set of analytically tractable univariate CRPS computations. MVG-CRPS provides explicit analytical gradients, enabling efficient integration into neural network training. The advantages of our approach are illustrated in Fig. 1, where the model trained with MVG-CRPS achieves higher accuracy while significantly reducing training time. The key contributions of our work are:

- We propose MVG-CRPS, a novel scoring rule for multivariate probabilistic forecasting that is less sensitive to outliers and extreme tails of the data distribution. Under the multivariate Gaussian family, we prove that MVG-CRPS is strictly proper.
- The proposed MVG-CRPS has a closed-form expression, allowing for the analytical computation of derivatives. This property facilitates efficient integration with backpropagation-based training in deep learning models and significantly reduces the computational cost compared to sampling-based alternatives.
- We perform extensive experiments with deep probabilistic forecasting models on real-world datasets. Our results demonstrate that MVG-CRPS balances accuracy and efficiency more effectively than standard scoring rules.

2 Related Work

2.1 Probabilistic Forecasting

Probabilistic forecasting focuses on modeling the complete probability distribution of target variables rather than producing single-point estimates. This comprehensive approach is essential for quantifying uncertainty inherent in time series data, thereby enabling more informed risk assessment and decision-making. Probabilistic forecasting methods typically fall into two main categories: parametric methods, which assume explicit probability density functions (PDFs), and non-parametric methods, which rely on quantile estimation [5].

Non-parametric methods generally forecast specific quantiles of the target distribution, thus avoiding restrictive parametric assumptions. A prominent example is the MQ-RNN [14], which leverages a Seq2Seq recurrent neural network (RNN) architecture to forecast multiple quantiles simultaneously. These quantile forecasts offer a robust approximation of the underlying distribution, making them particularly effective for capturing asymmetric and heavy-tailed behaviors.

Parametric methods assume a predefined probability distribution—such as Gaussian or Poisson—and estimate its parameters using neural networks. The DeepAR model [15], for instance, employs an RNN to capture hidden state transitions and predict Gaussian distribution parameters at each time step. GPVar [4], its multivariate extension, incorporates a Gaussian copula to transform observations

into Gaussian variables, thus modeling joint dependencies among multiple time series effectively. This method efficiently captures temporal and cross-series correlations through generalized least squares (GLS) approaches [16, 17] or dynamic regression [18].

Neural networks also facilitate modeling more complex probabilistic structures, including state-space models (SSMs) [19, 20], normalizing flows (NFs) [6], and diffusion models [21]. Additionally, copula-based methods explicitly model dependencies between multiple time series. Recent studies by Drouin et al. [22] and Ashok et al. [23] employ copulas to combine individual marginal distributions and dependency structures, achieving flexible multivariate modeling capabilities. Most existing approaches predominantly use the log-score as their optimization criterion.

2.2 Scoring Rules

Scoring rules quantitatively assess probabilistic forecast quality by comparing predicted distributions with observed outcomes. A scoring rule is deemed proper if it incentivizes honest forecasting, achieving its minimal expected score when the predicted distribution matches the true distribution. Formally, a scoring rule $s(p, q)$ is proper if the divergence $d(p, q) = s(p, q) - s(q, q)$ is non-negative and it is strictly proper if $d(p, q) = 0$ implies $p = q$ [24].

The negative log-likelihood (log-score) is a prevalent strictly proper scoring rule, evaluating predictive densities directly at observed outcomes. Widely adopted due to its analytical tractability, the log-score is particularly beneficial when the predictive density has a known parametric form [25]. The log-score is a strictly proper scoring rule and has several desirable properties, such as consistency and sensitivity to the entire distribution. In addition, the analytical tractability (closed-form expression and gradients for many distributions) makes it a convenient default in deep probabilistic forecasting models. However, for certain distributions (e.g., Gaussian), the log-score severely penalizes unlikely events, rendering it sensitive to outliers and extreme observations [26]. To mitigate this sensitivity, the CRPS provides a robust alternative in univariate contexts [27]. The CRPS quantifies discrepancies between the predictive cumulative distribution function (CDF) and observations, integrating absolute error over all potential thresholds. Unlike the exponential penalty in log-score, CRPS linearly penalizes deviations, thus reducing vulnerability to extreme events [28]. CRPS-based optimization techniques have demonstrated superior calibration and robustness compared to likelihood-based approaches in various probabilistic forecasting applications [28–30]. Minimum CRPS estimation specifically targets improved calibration by optimizing parameters directly to minimize CRPS rather than maximizing likelihood.

Multivariate forecasting introduces additional complexity due to inter-dependencies and higher dimensionality. While the log-score remains applicable, its sensitivity to outliers persists in this setting. The ES [9] generalizes the CRPS for multivariate distributions by computing expected distances between predictive and observed distributions. While ES effectively detects errors in the forecast mean, it is less sensitive to variance errors and, more critically, to misspecifications in the correlation structure among variables [31, 32]. The absence of a closed form expression also necessitates the use of Monte Carlo simulations to approximate the ES by drawing samples from the predictive distribution, which can be computationally expensive [see e.g., 33, 25, 13, 12].

To overcome the limited sensitivity of ES to the dependence structure, the variogram score (VS) was proposed by Scheuerer and Hamill [34]. VS explicitly targets inter-variable dependencies by comparing pairwise differences between forecasted and observed components. Similar to the ES, VS is typically approximated using ensemble forecasts or Monte Carlo sampling. However, it introduces additional computational complexity and still lacks a fully closed-form expression, limiting its direct applicability in large-scale or real-time settings. For a broader discussion of multivariate scoring rules and their properties, we refer readers to the comprehensive reviews by Gneiting and Katzfuss [35], Ziel and Berk [36], Waghmare and Ziegel [37] and Pic et al. [38].

The most relevant recent work is by Olafsdottir et al. [39], who propose a parameter estimation framework for multivariate spatial models by maximizing the average leave-one-out score (LOOS). Their method leverages the tractable conditionals of multivariate Gaussians and robust scoring rules like the CRPS. It is especially efficient for models with sparse precision matrices (e.g., Gaussian Markov random fields), but incurs notable overhead for general multivariate Gaussians due to the cost of computing all conditionals.

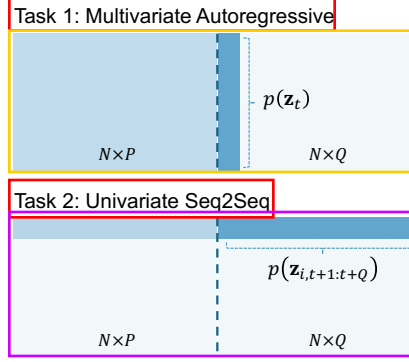


Figure 2: Illustration of the multivariate autoregressive and univariate Seq2Seq forecasting tasks.

3 Our Method

3.1 Multivariate Probabilistic Forecasting

Probabilistic forecasting aims to estimate the joint distribution over a collection of future quantities based on a given history of observations [35]. Denote the time series vector at a time point t as $\mathbf{z}_t = [z_{1,t}, \dots, z_{N,t}]^\top \in \mathbb{R}^N$, where N is the number of series. The problem of probabilistic forecasting can be formulated as $p(\mathbf{z}_{T+1:T+Q} | \mathbf{z}_{T-P+1:T}; \mathbf{x}_{T-P+1:T+Q})$, where $\mathbf{z}_{i,t:t_2} = [z_{i,t}, \dots, z_{i,t_2}]$, P is the conditioning range, Q is the prediction range, and T is the time point that splits the conditioning range and prediction range. $\mathbf{x}_t$ are some known covariates for both past and future time steps.

Multivariate probabilistic forecasting can be formulated in different ways. One way is over the time series dimension, where multiple interrelated variables are forecasted simultaneously at each time point. Considering an autoregressive model, where the predicted output is used as input for the next time step, this formulation can be factorized as

$$\begin{aligned} & p(\mathbf{z}_{T+1:T+Q} | \mathbf{z}_{T-P+1:T}; \mathbf{x}_{T-P+1:T+Q}) \\ &= \prod_{t=T+1}^{T+Q} p(\mathbf{z}_t | \mathbf{z}_{t-P:t-1}; \mathbf{x}_{t-P:t}) = \prod_{t=T+1}^{T+Q} p(\mathbf{z}_t | \mathbf{h}_t), \end{aligned} \quad (1)$$

where $\mathbf{h}_t$ is a state vector that encodes all the conditioning information used to generate the distribution parameters, typically via a neural network.

Another option is over the prediction horizon, where forecasts are made across multiple future time steps for one or more variables, capturing temporal dependencies and uncertainties over time. Considering a shared model across different series:

$$p(\mathbf{z}_{i,T+1:T+Q} | \mathbf{z}_{i,T-P+1:T}; \mathbf{x}_{i,T-P+1:T+Q}), \quad (2)$$

where $i = 1, \dots, N$ denotes the identifier of a particular time series. Since the model outputs forecasts for the entire prediction horizon directly, it is also called a Seq2Seq model. Without loss of generality, we use the first approach as an example to illustrate our method, since both approaches focus on estimating a multivariate distribution $p(\mathbf{z}_t)$ or $p(\mathbf{z}_{i,T+1:T+Q})$ (Fig. 2).

A typical probabilistic forecasting model assumes Gaussian noise; for example, it models $\mathbf{z}_t$ as jointly following a multivariate Gaussian distribution:

$$\mathbf{z}_t | \mathbf{h}_t \sim \mathcal{N}(\boldsymbol{\mu}(\mathbf{h}_t), \boldsymbol{\Sigma}(\mathbf{h}_t)), \quad (3)$$

where $\boldsymbol{\mu}(\cdot)$ and $\boldsymbol{\Sigma}(\cdot)$ are the functions mapping $\mathbf{h}_t$ to the mean and covariance parameters. The log-likelihood of the distribution given observed time series data up to time point T can be used as the loss function for optimizing a DL model:

$$\mathcal{L} = \sum_{t=1}^T \log p(\mathbf{z}_t | \theta(\mathbf{h}_t)) \propto \sum_{t=1}^T -\frac{1}{2} [\ln |\boldsymbol{\Sigma}_t| + \boldsymbol{\eta}_t^\top \boldsymbol{\Sigma}_t^{-1} \boldsymbol{\eta}_t], \quad (4)$$

where $\eta_t = \mathbf{z}_t - \boldsymbol{\mu}_t$. The above formulation simplifies to the univariate case when we set $N = 1$ for the model, with the same model being shared across all time series:

$$z_{i,t} \mid \mathbf{h}_{i,t} \sim \mathcal{N}(\mu(\mathbf{h}_{i,t}), \sigma^2(\mathbf{h}_{i,t})), \quad (5)$$

where $\mu(\cdot)$ and $\sigma(\cdot)$ map $\mathbf{h}_{i,t}$ to the mean and standard deviation of a Gaussian distribution. The corresponding log-likelihood becomes

$$\mathcal{L} = \sum_{t=1}^T \sum_{i=1}^N \log p(z_{i,t} \mid \theta(\mathbf{h}_{i,t})) \propto \sum_{t=1}^T \sum_{i=1}^N -\frac{1}{2} \epsilon_{i,t}^2 - \ln \sigma_{i,t}, \quad (6)$$

where $\epsilon_{i,t} = \frac{z_{i,t} - \mu_{i,t}}{\sigma_{i,t}}$. Eq. (4) and Eq. (6), when used as scoring rules to optimize the model, are generally referred to as the log-score and are widely employed in probabilistic forecasting.

For univariate problems, the CRPS is also a strictly proper scoring rule, defined as

$$\text{CRPS}(F, z) = \mathbb{E}_F |x - z| - \frac{1}{2} \mathbb{E}_F |x - x'|, \quad (7)$$

where F is the predictive CDF, z is the observation, and x and x' are independent random variables both associated with F . The CRPS has a closed-form expression when evaluating a Gaussian-distributed variable $z \sim \mathcal{N}(\mu, \sigma^2)$ [28]:

$$\text{CRPS}(\Phi, z) = z(2\Phi(z) - 1) + 2\varphi(z) - \frac{1}{\sqrt{\pi}}, \quad (8)$$

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$$\text{CRPS}(F_{\mu,\sigma}, z) = \sigma \text{CRPS}\left(\Phi, \frac{z - \mu}{\sigma}\right), \quad (9)$$

where $F_{\mu,\sigma}(z) = \Phi\left(\frac{z - \mu}{\sigma}\right)$, Φ and φ are the CDF and PDF of the standard Gaussian distribution.

The CRPS has been shown to be a more robust alternative to the log-score as a loss function in many problems [28, 27, 40]. We observe that the log-score can grow arbitrarily large in magnitude when a single outlier disproportionately influences the loss function, owing to the unbounded nature of the logarithmic function (Eq. (4) and Eq. (6)). Additionally, the quadratic form of the error terms in the Gaussian likelihood also makes it sensitive to outliers (e.g., $\epsilon_{i,t}^2$ in Eq. (6)). In contrast, the CRPS evaluates the entire predictive distribution rather than concentrating solely on the likelihood of individual data points (Eq. (8)). Moreover, the CRPS can directly replace the log-score, providing analytical gradients with respect to μ and σ for backpropagation. However, for a multivariate Gaussian distribution, the CRPS does not have a widely used closed-form expression.

3.2 MVG-CRPS as Loss Function for Multivariate Forecasting

In multivariate probabilistic forecasting, proper scoring rules such as the log-score (Eq. (4)) and the ES are used to evaluate predictive performance. The ES generalizes the CRPS to assess probabilistic forecasts of vector-valued random variables [9]:

$$\text{ES}(F, \mathbf{z}) = \mathbb{E}_{\mathbf{x} \sim F} \|\mathbf{x} - \mathbf{z}\|^\beta - \frac{1}{2} \mathbb{E}_{\mathbf{x}, \mathbf{x}' \sim F} \|\mathbf{x} - \mathbf{x}'\|^\beta, \quad (10)$$

where $\|\cdot\|$ denotes the Euclidean norm and $\beta = 1$ is commonly used in the literature [23]. With $\beta = 1$, the ES essentially becomes a multivariate extension of the CRPS and grows linearly with respect to the norm, making it less sensitive to outliers compared to the log-score. Since there is no simple closed-form expression for Eq. (10), it is often approximated using Monte Carlo methods, where multiple samples $\{\mathbf{x}_i\}_{i=1}^n$ are drawn from the forecast distribution to approximate the expected values:

$$\text{ES}(F, \mathbf{z}) = \frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i - \mathbf{z}\|^\beta - \frac{1}{2n^2} \sum_{i=1}^n \sum_{j=1}^n \|\mathbf{x}_i - \mathbf{x}_j\|^\beta. \quad (11)$$

However, a significant disadvantage of using Eq. (11) as the loss function is that it requires Monte Carlo sampling during the training process, which can substantially slow down training and create noisy gradients.

In this section, we propose the MVG-CRPS, a robust and efficient loss function designed as an alternative for multivariate forecasting. This loss function grows linearly with the prediction error, making it more robust than the log-score. Additionally, it does not require sampling during the training process, rendering it more efficient than the ES.

Our proposed method is based on the whitening transformation of a time series vector that follows a multivariate Gaussian distribution, $\mathbf{z}_t \sim \mathcal{N}(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$. The whitening process transforms a random vector with a known covariance matrix into a new random vector whose covariance matrix is the identity matrix. As a result, the elements of the transformed vector have unit variance and are uncorrelated. This transformation begins by performing the singular value decomposition (SVD) of the covariance matrix:

$$\boldsymbol{\Sigma}_t = \mathbf{U}_t \mathbf{S}_t \mathbf{U}_t^\top, \quad (12)$$

where $\mathbf{S}_t = \text{diag}([\lambda_{1,t}, \dots, \lambda_{N,t}]^\top)$ is a diagonal matrix containing the eigenvalues of $\boldsymbol{\Sigma}_t$, and $\mathbf{U}_t$ is the orthonormal matrix of corresponding eigenvectors. We then define

$$\mathbf{v}_t = \mathbf{U}_t^\top (\mathbf{z}_t - \boldsymbol{\mu}_t), \quad (13)$$

where $\mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{S}_t)$ is a random vector with a uncorrelated multivariate Gaussian distribution, having variances λ_i (i.e., the corresponding eigenvalue) along the diagonal of its covariance matrix. Next, we define

$$\mathbf{w}_t = \mathbf{S}_t^{-\frac{1}{2}} \mathbf{v}_t = \mathbf{S}_t^{-\frac{1}{2}} \mathbf{U}_t^\top (\mathbf{z}_t - \boldsymbol{\mu}_t), \quad (14)$$

where $\mathbf{w}_t$ is a random vector with each element following a standard Gaussian distribution, i.e., $w_{i,t} \sim \mathcal{N}(0, 1)$. We can then apply Eq. (8) individually to each element and formulate the MVG-CRPS mimicking Eq. (9) for multivariate problem:

$$\text{MCRPS}(\Phi_N(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t), \mathbf{z}_t) = \sum_{i=1}^N \text{CRPS}(\Phi(0, \lambda_{i,t}), v_{i,t}) = \sum_{i=1}^N \sqrt{\lambda_{i,t}} \text{CRPS}(\Phi, w_{i,t}), \quad (15)$$

where $\Phi_N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is the CDF of multivariate Gaussian with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$.

The overall loss function for training the model is then formulated over an observation period T :

$$\mathcal{L} = \sum_{t=1}^T \text{MCRPS}(\Phi_N(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t), \mathbf{z}_t). \quad (16)$$

By leveraging PCA whitening, the MVG-CRPS effectively discriminates between differences in both the mean and covariance within the multivariate Gaussian family—whereas the ES may overlook subtle covariance discrepancies and the log-score lacks robustness. The key advantage of MVG-CRPS lies in its ability to exploit the closed-form expression of the univariate CRPS by decorrelating multivariate time series variables via PCA whitening. This transformation enables the evaluation of marginal distributions in an orthogonalized space, where the whitening is derived from the original covariance matrix. As a result, the optimization process preserves and is sensitive to the dependence structure of the original multivariate distribution. Under the Gaussian assumption, MVG-CRPS constitutes a strictly proper scoring rule (see Appendix [§A](#)).

4 Experiments

4.1 Datasets and Models

We evaluate MVG-CRPS on two forecasting tasks: multivariate autoregressive forecasting using the RNN-based GPVar [4] and a decoder-only Transformer [41], and univariate Seq2Seq forecasting using the MLP-based N-HITS model [42].

To generate the distribution parameters for probabilistic forecasting, we employ a Gaussian distribution head based on the hidden state $\mathbf{h}_t$ produced by the model. Specifically, for the multivariate autoregressive forecasting, following Salinas et al. [4], we parameterize the mean vector as $\boldsymbol{\mu}(\mathbf{h}_t) = [\mu_1(\mathbf{h}_{1,t}), \dots, \mu_N(\mathbf{h}_{N,t})]^\top \in \mathbb{R}^N$ and adopt a low-rank-plus-diagonal parameterization of the covariance matrix $\boldsymbol{\Sigma}(\mathbf{h}_t) = \mathbf{L}_t \mathbf{L}_t^\top + \text{diag}(\mathbf{d}_t)$, where $\mathbf{d}_t = [d_1(\mathbf{h}_{1,t}), \dots, d_N(\mathbf{h}_{N,t})]^\top \in \mathbb{R}_+^N$ and $\mathbf{L}_t = [\mathbf{l}_1(\mathbf{h}_{1,t}), \dots, \mathbf{l}_N(\mathbf{h}_{N,t})]^\top \in \mathbb{R}^{N \times R}$, $R \ll N$ is the rank parameter. Here, $\mu_i(\cdot)$, $d_i(\cdot)$,

and $\mathbf{l}_i(\cdot)$ are the mapping functions that generate the mean and covariance parameters for each time series i based on the hidden state $\mathbf{h}_{i=1:N,t}$. This parameterization guarantees that $\Sigma(\mathbf{h}_t)$ is full-rank, ensuring that the eigen-decomposition in Eq. (12) is always well-defined. In practice, we use shared mapping functions across all time series, denoted as $\mu_i = \tilde{\mu}$, $d_i = \tilde{d}$, and $\mathbf{l}_i = \mathbf{l}$. This parameterization ensures that $\Sigma(\mathbf{h}_t)$ is positive definite and efficiently parameterized. The diagonal component provides stability, while the low-rank component captures the covariance structure. The Gaussian assumption also enables the use of random subsets of time series (i.e., batch size $B < N$) for model optimization in each iteration, making it feasible to apply our method to high-dimensional time series datasets. Similarly, in the univariate Seq2Seq forecasting task, the mean $\mu(\mathbf{h}_t)$ and covariance $\Sigma(\mathbf{h}_t)$ are defined over the forecast horizon for each specific time series, based on the hidden states $\mathbf{h}_{i,t-T+1:T+Q}$. As a result, we can model the joint distribution $p(\mathbf{z}_{i,T+1:T+Q})$ over the forecasted values. We implemented our models using PyTorch Forecasting [43], with input data consisting of lagged time series values and covariates. Extensive experiments were conducted on a variety of real-world time series datasets from GluonTS [44] (see Appendix §B). Full details of the experimental setup are provided in Appendix §C.

4.2 Toy Example

We first perform a toy experiment following Roordink and Hess [45] using a true distribution $P = \mathcal{N}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 & 0.8 \\ 0.8 & 4 \end{bmatrix}\right)$ and a predictive distribution $Q = \mathcal{N}\left(\begin{bmatrix} \mu \\ -1 \end{bmatrix}, \begin{bmatrix} \sigma^2 & 2\rho\sigma \\ 2\rho\sigma & 4 \end{bmatrix}\right)$ where we control the deviation of the three parameters μ, ρ, σ to study the various properties of different scores. As shown in Fig. 3, the log-score increases sharply when the standard deviation σ or correlation coefficient ρ deviate from their true values, indicating high sensitivity to covariance misspecification. The ES shows lower sensitivity to the covariance structure but produces non-smooth curves due to its sample-based approximation. In contrast, the MVG-CRPS displays comparable sensitivity to deviations in all three parameters. It also produces smooth curves with a clear minimum at zero deviation, reflecting its closed-form evaluation.

We further examine the robustness of different scoring rules for estimating the parameters of this predictive distribution under data contamination, and analyze the trade-off between computational cost and estimation accuracy for the ES with varying sample sizes (see Appendix §D.1). Overall, MVG-CRPS demonstrates greater robustness than the log-score across all three parameters, particularly for μ and σ , and provides more consistent estimates than the ES due to its sampling-free formulation (Fig. A1). We also observe that the ES produces less accurate estimates than MVG-CRPS for μ and σ . Although we do not claim superiority over the ES beyond efficiency, this discrepancy is likely attributable to the variance introduced by its Monte Carlo approximation. Additionally, we observe that the gains in estimation accuracy diminish rapidly as the sample size increases, and the ES does not significantly outperform MVG-CRPS even with 1,000 samples (Fig. A2). Meanwhile, the computational cost of the ES increases monotonically with sample size.

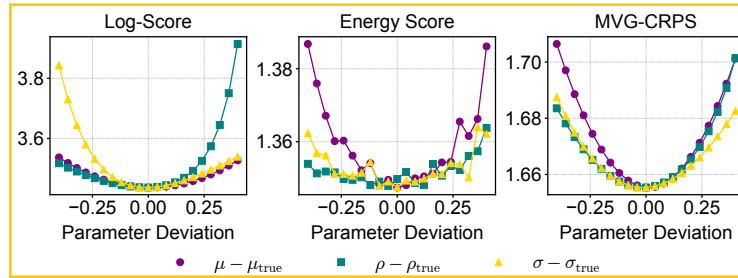


Figure 3: Sensitivity of scoring rules to parameter deviations in the predicted mean, standard deviation, and correlation coefficient from the true data distribution $\mu_{\text{true}} = 1, \sigma_{\text{true}} = 1, \rho_{\text{true}} = 0.4$. The ES values are computed with a sample size of 500.

4.3 Quantitative Evaluation

We evaluate the MVG-CRPS against models trained with the log-score and the ES using three common metrics for probabilistic forecasts: CRPS_{sum} , $\text{CRPS}_{\text{mean}}$, and the ES (see Appendix §C.5).

Table 1: Comparison of CRPS_{sum} across different scoring rules in the multivariate autoregressive forecasting task. The best scores are in boldface. MVG-CRPS scores are underlined when they are not the best overall but exceed the log-score.

	VAR	GPVar			Transformer		
		log-score	energy score	MVG-CRPS	log-score	energy score	MVG-CRPS
elec_au	N/A	0.1261±0.0009	0.0887±0.0004	0.0967±0.0008	0.1633±0.0005	0.1492±0.0006	0.0793±0.0004
cif_2016	1.0000±0.0000	0.0122±0.0004	0.0420±0.0006	0.0111±0.0005	0.0118±0.0003	0.0240±0.0014	0.0107±0.0002
electricity	0.1315±0.0006	0.0419±0.0008	0.0616±0.0004	0.0249±0.0006	0.0362±0.0002	0.0368±0.0004	0.0294±0.0004
elec_weekly	0.1126±0.0011	0.1515±0.0028	0.0417±0.0014	0.0772±0.0031	0.0937±0.0026	0.0403±0.0013	0.0448±0.0014
exchange_rate	0.0033±0.0000	0.0207±0.0004	0.0030±0.0001	0.0041±0.0001	0.0047±0.0003	0.0067±0.0003	0.0091±0.0004
kdd_cup	N/A	0.3743±0.0019	0.3210±0.0019	0.2358±0.0014	0.2076±0.0013	0.4789±0.0030	0.1959±0.0017
m1_yearly	N/A	0.4397±0.0041	0.4801±0.0022	0.3566±0.0029	0.5344±0.0109	0.3291±0.0047	0.4563±0.0111
m3_yearly	N/A	0.3607±0.0084	0.2186±0.0042	0.1423±0.0053	0.3156±0.0102	0.4050±0.0061	0.2325±0.0094
nn5_daily	0.2303±0.0005	0.0998±0.0004	0.0958±0.0003	0.0948±0.0003	0.0991±0.0003	0.0883±0.0004	0.0811±0.0002
saugeenday	N/A	0.4040±0.0047	0.3733±0.0048	0.3941±0.0055	0.3771±0.0088	0.3689±0.0053	0.3705±0.0047
sunspot	N/A	18.7115±1.3296	23.3988±0.9662	17.2438±0.5833	39.7454±1.4841	16.6556±0.6167	22.6495±0.6752
tourism	0.1394±0.0012	0.2217±0.0027	0.2112±0.0014	0.2004±0.0022	0.2100±0.0017	0.2087±0.0020	0.2082±0.0015
traffic	3.5241±0.0084	0.0742±0.0004	0.0505±0.0002	0.0868±0.0002	0.0658±0.0002	0.0667±0.0002	0.0683±0.0000
Avg. Rank		2.62	1.92	1.46	2.38	2.00	1.62

Table 2: Training time (in minutes) for GPVar using different scoring rules in the multivariate autoregressive forecasting task. Reported times include early stopping and reflect differences in convergence speed across loss functions.

	log-score		energy score		MVG-CRPS	
	per epoch	total	per epoch	total	per epoch	total
elec_au	0.86	33.53	16.29	717.00	0.78	29.14
cif_2016	0.13	1.58	4.83	401.04	0.12	3.85
electricity	0.40	67.38	11.17	782.40	0.38	22.70
elec_weekly	0.30	14.61	10.95	383.52	0.26	18.77
exchange_rate	0.25	16.40	10.20	663.60	0.29	23.63
kdd_cup	0.42	11.32	14.23	2063.52	0.42	28.79
m1_yearly	0.19	3.71	5.66	469.92	0.18	8.02
m3_yearly	0.43	7.30	10.80	291.72	0.42	14.49
nn5_daily	0.29	9.21	11.64	244.50	0.27	14.53
saugeenday	0.23	12.65	10.70	524.46	0.15	15.32
sunspot	0.44	26.85	10.73	397.26	0.42	16.96
tourism	0.49	23.96	10.56	243.00	0.46	12.51
traffic	0.94	76.98	14.92	1044.60	0.92	92.46

for definitions). Table 1 presents a comparison of CRPS_{sum} for the multivariate autoregressive forecasting task. Overall, the MVG-CRPS achieves the best average rank among the three scoring rules. Notably, it consistently outperforms the log-score across most datasets, indicating that MVG-CRPS leads to models with higher-quality forecasts. As shown in later sections, this improvement is attributed to MVG-CRPS being less sensitive to outliers. Compared to the ES, MVG-CRPS achieves comparable or better performance (Table 1) while being more efficient during training (Table 2). It is important to note that we do not claim MVG-CRPS is more robust than ES; rather, our focus is on its efficiency compared to ES. Results for CRPS_{mean} and the ES are provided in Appendix §D.2, and results for the univariate Seq2Seq forecasting task are presented in Appendix §D.3. In both tasks, MVG-CRPS achieves consistent performance across all three evaluation metrics.

4.4 Qualitative Evaluation

To illustrate the robustness of MVG-CRPS, we compare the output covariance matrices from models trained with different loss functions and visualize their probabilistic forecasts. In Fig. 4, the log-score model produces covariance matrices that occasionally exhibit large covariances, despite normalization applied to each time series. This behavior likely reflects the influence of large tail errors during training. In contrast, the MVG-CRPS model captures similar covariance patterns without extreme values, indicating improved robustness to outliers. To highlight the practical impact, we compare GPVar forecasts on the electricity dataset (Fig. 5). MVG-CRPS yields sharper and better-calibrated predictions, while the log-score model occasionally produces overly wide intervals,

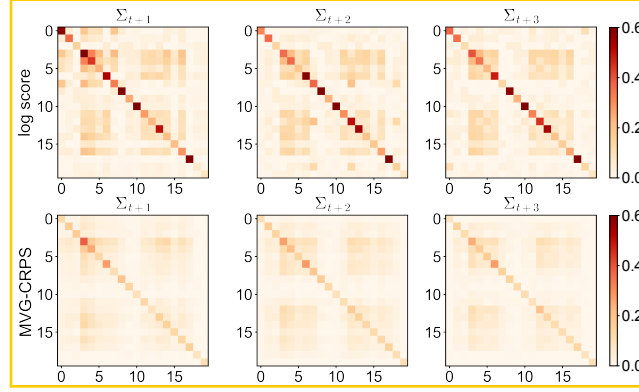


Figure 4: Comparison of output covariance matrices Σ_t from GPVar on the elec_weekly dataset. For visual clarity, covariance values are clipped between 0 and 0.6.

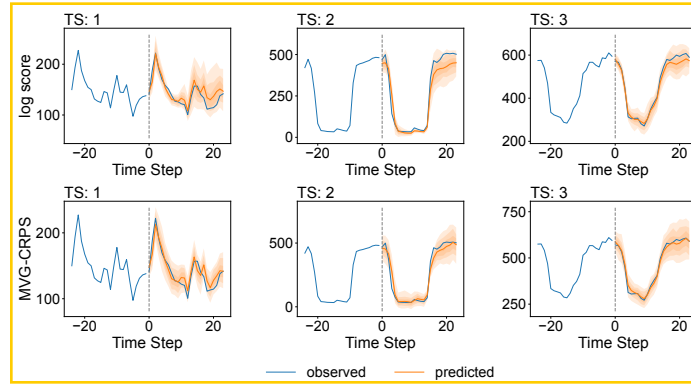


Figure 5: Comparison of probabilistic forecasts from GPVar on the electricity dataset.

reflecting greater sensitivity to outliers (e.g., TS 1). Results for the univariate Seq2Seq forecasting task are provided in Appendix D.3.

5 Conclusion

This paper introduced the MVG-CRPS, a novel strictly proper scoring rule specifically designed for multivariate Gaussian probabilistic forecasting. MVG-CRPS addresses the sensitivity of the log-score to outliers and overcomes the computational inefficiency inherent to the ES. By applying a whitening transformation and leveraging the closed-form expression of the univariate CRPS, our approach achieves robustness to extreme values while remaining computationally efficient and easily integrable into deep learning frameworks. Moreover, the MVG-CRPS exhibits high sensitivity to both the mean and covariance of the predictive distribution—comparable to the log-score—while preserving the robustness properties of the ES. Empirical evaluations on real-world datasets demonstrated significant improvements in both predictive accuracy and robustness compared to existing scoring rules.

Beyond forecasting, the general formulation of MVG-CRPS extends naturally to broader probabilistic regression contexts, such as robust Gaussian process regression, by replacing conventional negative marginal likelihood objectives. Future directions include leveraging copula transformations to extend the MVG-CRPS to non-Gaussian distributions and exploring more efficient covariance parameterizations to enhance scalability. Currently, scalability remains constrained by the computational demands of eigen-decomposition in large-batch scenarios. A possible solution to mitigate this limitation is to adopt an isotropic noise parameterization, i.e., $\Sigma = LL^T + \sigma^2 \mathbf{I}$, which enables more efficient computation of the SVD.

References

- [1] Jan JJ Groen, Richard Paap, and Francesco Ravazzolo. Real-time inflation forecasting in a changing world. *Journal of Business & Economic Statistics*, 31(1):29–44, 2013.
- [2] TN Palmer. Towards the probabilistic earth-system simulator: A vision for the future of climate and weather prediction. *Quarterly Journal of the Royal Meteorological Society*, 138(665):841–861, 2012.
- [3] Hayley E Jones and David J Spiegelhalter. Improved probabilistic prediction of healthcare performance indicators using bidirectional smoothing models. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 175(3):729–747, 2012.
- [4] David Salinas, Michael Bohlke-Schneider, Laurent Callot, Roberto Medico, and Jan Gasthaus. High-dimensional multivariate forecasting with low-rank gaussian copula processes. *Advances in Neural Information Processing Systems*, 32, 2019.
- [5] Konstantinos Benidis, Syama Sundar Rangapuram, Valentin Flunkert, Yuyang Wang, Danielle Maddix, Caner Turkmen, Jan Gasthaus, Michael Bohlke-Schneider, David Salinas, Lorenzo Stella, et al. Deep learning for time series forecasting: Tutorial and literature survey. *ACM Computing Surveys*, 55(6):1–36, 2022.
- [6] Kashif Rasul, Abdul-Saboor Sheikh, Ingmar Schuster, Urs Bergmann, and Roland Vollgraf. Multivariate probabilistic time series forecasting via conditioned normalizing flows. In *International Conference on Learning Representations*, 2021.
- [7] Manuel Gebetsberger, Jakob W Messner, Georg J Mayr, and Achim Zeileis. Estimation methods for nonhomogeneous regression models: Minimum continuous ranked probability score versus maximum likelihood. *Monthly Weather Review*, 146(12):4323–4338, 2018.
- [8] Mathias Blicher Bierregård, Jan Kloppenborg Møller, and Henrik Madsen. An introduction to multivariate probabilistic forecast evaluation. *Energy and AI*, 4:100058, 2021.
- [9] Tilmann Gneiting and Adrian E Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- [10] James E Matheson and Robert L Winkler. Scoring rules for continuous probability distributions. *Management Science*, 22(10):1087–1096, 1976.
- [11] Tilmann Gneiting and Adrian E Raftery. Weather forecasting with ensemble methods. *Science*, 310(5746):248–249, 2005.
- [12] Lorenzo Pacchiardi, Rilwan A Adewoyin, Peter Dueben, and Ritabrata Dutta. Probabilistic forecasting with generative networks via scoring rule minimization. *Journal of Machine Learning Research*, 25(45):1–64, 2024.
- [13] Jieyu Chen, Tim Janke, Florian Steinke, and Sebastian Lerch. Generative machine learning methods for multivariate ensemble postprocessing. *The Annals of Applied Statistics*, 18(1):159–183, 2024.
- [14] Ruofeng Wen, Kari Torkkola, Balakrishnan Narayanaswamy, and Dhruv Madeka. A multi-horizon quantile recurrent forecaster. *arXiv preprint arXiv:1711.11053*, 2017.
- [15] David Salinas, Valentin Flunkert, Jan Gasthaus, and Tim Januschowski. Deepar: Probabilistic forecasting with autoregressive recurrent networks. *International Journal of Forecasting*, 36(3):1181–1191, 2020.
- [16] Vincent Zhihao Zheng, Seongjin Choi, and Lijun Sun. Better batch for deep probabilistic time series forecasting. In *International Conference on Artificial Intelligence and Statistics*, pages 91–99, 2024.
- [17] Vincent Zhihao Zheng and Lijun Sun. Multivariate probabilistic time series forecasting with correlated errors. *Advances in Neural Information Processing Systems*, 37, 2024.
- [18] Vincent Zhihao Zheng, Seongjin Choi, and Lijun Sun. Probabilistic traffic forecasting with dynamic regression. *Transportation Science*, 2025.
- [19] Syama Sundar Rangapuram, Matthias W Seeger, Jan Gasthaus, Lorenzo Stella, Yuyang Wang, and Tim Januschowski. Deep state space models for time series forecasting. *Advances in Neural Information Processing Systems*, 31, 2018.

- [20] Emmanuel de Bézenac, Syama Sundar Rangapuram, Konstantinos Benidis, Michael Bohlke-Schneider, Richard Kurlle, Lorenzo Stella, Hilaf Hasson, Patrick Gallinari, and Tim Januschowski. Normalizing kalman filters for multivariate time series analysis. *Advances in Neural Information Processing Systems*, 33:2995–3007, 2020.
- [21] Kashif Rasul, Calvin Seward, Ingmar Schuster, and Roland Vollgraf. Autoregressive denoising diffusion models for multivariate probabilistic time series forecasting. In *International Conference on Machine Learning*, pages 8857–8868, 2021.
- [22] Alexandre Drouin, Étienne Marcotte, and Nicolas Chapados. Tactis: Transformer-attentional copulas for time series. In *International Conference on Machine Learning*, pages 5447–5493, 2022.
- [23] Arjun Ashok, Étienne Marcotte, Valentina Zantedeschi, Nicolas Chapados, and Alexandre Drouin. Tactis-2: Better, faster, simpler attentional copulas for multivariate time series. In *International Conference on Learning Representations*, 2024.
- [24] Jochen Bröcker. Reliability, sufficiency, and the decomposition of proper scores. *Quarterly Journal of the Royal Meteorological Society: A journal of the atmospheric sciences, applied meteorology and physical oceanography*, 135(643):1512–1519, 2009.
- [25] Anastasios Panagiotelis, Puwasala Gamakumara, George Athanasopoulos, and Rob J Hyndman. Probabilistic forecast reconciliation: Properties, evaluation and score optimisation. *European Journal of Operational Research*, 306(2):693–706, 2023.
- [26] Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E Raftery. Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 69(2):243–268, 2007.
- [27] Stephan Rasp and Sebastian Lerch. Neural networks for postprocessing ensemble weather forecasts. *Monthly Weather Review*, 146(11):3885–3900, 2018.
- [28] Tilmann Gneiting, Adrian E Raftery, Anton H Westveld, and Tom Goldman. Calibrated probabilistic forecasting using ensemble model output statistics and minimum crps estimation. *Monthly Weather Review*, 133(5):1098–1118, 2005.
- [29] Kin G Olivares, Geoffrey Négier, Ruijun Ma, O Nangba Meetei, Mengfei Cao, and Michael W Mahoney. Probabilistic forecasting with coherent aggregation. *arXiv preprint arXiv:2307.09797*, 2023.
- [30] Simon Lang, Mihai Alexe, Mariana CA Clare, Christopher Roberts, Rilwan Adewoyin, Zied Ben Boual-lègue, Matthew Chantry, Jesper Dramsch, Peter D Dueben, Sara Hahner, et al. Aifs-crps: Ensemble forecasting using a model trained with a loss function based on the continuous ranked probability score. *arXiv preprint arXiv:2412.15832*, 2024.
- [31] Pierre Pinson and Julija Tastu. Discrimination ability of the energy score. 2013.
- [32] Carol Alexander, Michael Coulon, Yang Han, and Xiaochun Meng. Evaluating the discrimination ability of proper multi-variate scoring rules. *Annals of Operations Research*, 334(1):857–883, 2024.
- [33] Diane Bouchacourt, Pawan K Mudigonda, and Sebastian Nowozin. Disco nets: Dissimilarity coefficients networks. *Advances in Neural Information Processing Systems*, 29, 2016.
- [34] Michael Scheuerer and Thomas M Hamill. Variogram-based proper scoring rules for probabilistic forecasts of multivariate quantities. *Monthly Weather Review*, 143(4):1321–1334, 2015.
- [35] Tilmann Gneiting and Matthias Katzfuss. Probabilistic forecasting. *Annual Review of Statistics and Its Application*, 1(1):125–151, 2014.
- [36] Florian Ziel and Kevin Berk. Multivariate forecasting evaluation: On sensitive and strictly proper scoring rules. *arXiv preprint arXiv:1910.07325*, 2019.
- [37] Kartik Waghmare and Johanna Ziegel. Proper scoring rules for estimation and forecast evaluation. *arXiv preprint arXiv:2504.01781*, 2025.
- [38] Romain Pic, Clément Dombry, Philippe Naveau, and Maxime Taillardat. Proper scoring rules for multivariate probabilistic forecasts based on aggregation and transformation. *Advances in Statistical Climatology, Meteorology and Oceanography*, 11(1):23–58, 2025.
- [39] Helga Kristin Olafsdottir, Holger Rootzén, and David Bolin. Fast and robust cross-validation-based scoring rule inference for spatial statistics. *arXiv preprint arXiv:2408.11994*, 2024.

- [40] Abdulmajid Murad, Frank Alexander Kraemer, Kerstin Bach, and Gavin Taylor. Probabilistic deep learning to quantify uncertainty in air quality forecasting. *Sensors*, 21(23):8009, 2021.
- [41] Alec Radford, Karthik Narasimhan, Tim Salimans, Ilya Sutskever, et al. Improving language understanding by generative pre-training. 2018.
- [42] Cristian Challu, Kin G Olivares, Boris N Oreshkin, Federico Garza Ramirez, Max Mergenthaler Canseco, and Artur Dubrawski. Nhits: Neural hierarchical interpolation for time series forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 6989–6997, 2023.
- [43] Jan Beitner. Pytorch forecasting. <https://pytorch-forecasting.readthedocs.io>, 2020.
- [44] Alexander Alexandrov, Konstantinos Benidis, Michael Bohlke-Schneider, Valentin Flunkert, Jan Gasthaus, Tim Januschowski, Danielle C Maddix, Syama Rangapuram, David Salinas, Jasper Schulz, et al. Gluonts: Probabilistic and neural time series modeling in python. *The Journal of Machine Learning Research*, 21(1):4629–4634, 2020.
- [45] Daan Roordink and Sibylle Hess. Scoring rule nets: Beyond mean target prediction in multivariate regression. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pages 190–205. Springer, 2023.
- [46] Alfred Horn. Doubly stochastic matrices and the diagonal of a rotation matrix. *American Journal of Mathematics*, 76(3):620–630, 1954.
- [47] Roger A Horn and Charles R Johnson. *Matrix analysis*. Cambridge University Press, 2012.
- [48] Taesung Kim, Jinhee Kim, Yunwon Tae, Cheonbok Park, Jang-Ho Choi, and Jaegul Choo. Reversible instance normalization for accurate time-series forecasting against distribution shift. In *International Conference on Learning Representations*, 2021.
- [49] Helmut Lütkepohl. *New Introduction to Multiple Time Series Analysis*. Springer Science & Business Media, 2005.

Appendix

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A MVG-CRPS is Strictly Proper

Theorem A.1. Let $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p)$ be a true N -variate Gaussian distribution where the covariance admits eigen-decomposition $\boldsymbol{\Sigma}_p = \mathbf{U}_p \mathbf{S}_p \mathbf{U}_p^\top$, with $\mathbf{S}_p = \text{diag}(\boldsymbol{\lambda}_p)$ containing nonincreasing eigenvalues $\boldsymbol{\lambda}_p = [\lambda_1^p, \dots, \lambda_N^p]^\top$ and $\mathbf{U}_p$ being the corresponding orthonormal matrix. Consider a predictive Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}_q, \boldsymbol{\Sigma}_q)$ where covariance $\boldsymbol{\Sigma}_q$ admits the eigen-decomposition $\boldsymbol{\Sigma}_q = \mathbf{U}_q \mathbf{S}_q \mathbf{U}_q^\top$ with $\mathbf{S}_q = \text{diag}(\boldsymbol{\lambda}_q)$. Define the transformed variable $\mathbf{v} = \mathbf{U}_q^\top (\mathbf{z} - \boldsymbol{\mu}_q) = [v_1, \dots, v_N]^\top$. The proposed MVG-CRPS

$$\text{MCRPS}(\Phi_N(\boldsymbol{\mu}_q, \boldsymbol{\Sigma}_q), \mathbf{z}) = \sum_{i=1}^N \text{CRPS}(\Phi(0, \lambda_i^q), v_i)$$

is proper and strictly proper for multivariate Gaussian distributions.

Proof. Given that $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p)$, we have the transformed variable $\mathbf{v} \sim \mathcal{N}(\boldsymbol{\mu}_v, \boldsymbol{\Sigma}_v)$ with $\boldsymbol{\mu}_v = \mathbf{U}_q^\top (\boldsymbol{\mu}_p - \boldsymbol{\mu}_q) = [\nu_1, \dots, \nu_N]^\top$ and $\boldsymbol{\Sigma}_v = \mathbf{U}_q^\top \boldsymbol{\Sigma}_p \mathbf{U}_q = \mathbf{U}_q^\top \mathbf{U}_p \mathbf{S}_p \mathbf{U}_p^\top \mathbf{U}_q = \mathbf{U}_v \mathbf{S}_p \mathbf{U}_v^\top$, where $\mathbf{U}_v = \mathbf{U}_q^\top \mathbf{U}_p$ is an orthonormal matrix. Thus, each v_i has a marginal distribution $v_i \sim \mathcal{N}(\nu_i, \tau_i)$ for $i = 1, \dots, N$, with $\boldsymbol{\tau} = \text{diag}(\boldsymbol{\Sigma}_v) = \text{diag}(\mathbf{U}_v \mathbf{S}_p \mathbf{U}_v^\top) = [\tau_1, \dots, \tau_N]^\top$. Taking the expectation

of MCRPS $(\Phi_N(\mu_q, \Sigma_q), \mathbf{z})$ under the true distribution, we have

$$\begin{aligned}
\mathbb{E}_{\mathbf{z} \sim \mathcal{N}(\mu_p, \Sigma_p)} [\text{MCRPS}(\Phi_N(\mu_q, \Sigma_q), \mathbf{z})] &= \sum_{i=1}^N \mathbb{E}_{v_i \sim \mathcal{N}(\nu_i, \tau_i)} [\text{CRPS}(\Phi(0, \lambda_i^q), v_i)] \\
&\geq \sum_{i=1}^N \mathbb{E}_{v_i \sim \mathcal{N}(\nu_i, \tau_i)} [\text{CRPS}(\Phi(\nu_i, \tau_i), v_i)] \\
&= \sum_{i=1}^N \mathbb{E}_{\eta_i \sim \mathcal{N}(0, \tau_i)} [\text{CRPS}(\Phi(0, \tau_i), \eta_i)] \\
&= \mathbb{E}_{v \sim \mathcal{N}(0, 1)} [\text{CRPS}(\Phi, v)] \times \sum_{i=1}^N \sqrt{\tau_i} \\
&\geq \mathbb{E}_{v \sim \mathcal{N}(0, 1)} [\text{CRPS}(\Phi, v)] \times \sum_{i=1}^N \sqrt{\lambda_i^p} \\
&= \mathbb{E}_{\mathbf{z} \sim \mathcal{N}(\mu_p, \Sigma_p)} [\text{MCRPS}(\Phi_N(\mu_p, \Sigma_p), \mathbf{z})].
\end{aligned} \tag{17}$$

The first inequality is a direct result of CRPS being a strictly proper scoring rule for univariate Gaussian distributions. We now prove the second inequality.

Recall that $\boldsymbol{\tau} = \text{diag}(\Sigma_p)$ and $\Sigma_p = \mathbf{U}_q^\top \Sigma_p \mathbf{U}_q$. Let $\boldsymbol{\tau}^*$ be the monotone nonincreasing rearrangement of $\boldsymbol{\tau}$. By the Schur-Horn theorem [46], the diagonal vector $\boldsymbol{\tau}^*$ is majorized by the eigenvalues λ_p^p :

$$\sum_{i=1}^k \tau_i^* \leq \sum_{i=1}^k \lambda_i^p,$$

for $k = 1, 2, \dots, N-1$, and

$$\sum_{i=1}^N \tau_i^* = \sum_{i=1}^N \lambda_i^p.$$

Since $f(x) = \sqrt{x}$ is a concave function, Karamata's majorization inequality yields

$$\sum_{i=1}^N \sqrt{\lambda_i^p} \leq \sum_{i=1}^N \sqrt{\tau_i^*} = \sum_{i=1}^N \sqrt{\tau_i}, \tag{18}$$

which proves the second inequality in Eq. (17). Hence, the MVG-CRPS is a proper scoring rule for the multivariate Gaussian distribution.

Equality in Eq. (18) is obtained if, for every i , $\tau_i^* = \lambda_i^p$. By the Schur-Horn theorem, this forces Σ_p to be a diagonal matrix (Theorem 4.3.45 in Horn and Johnson [47]). Meanwhile, the CRPS inequality in Eq. (17) is tight exactly when, for every i , $\nu_i = 0$ and $\tau_i = \lambda_i^q$, implying that $\mathbf{U}_q^\top (\mu_p - \mu_q) = \mathbf{0}$ and $\text{diag}(\Sigma_p) = \text{diag}(\Sigma_q)$. Since Σ_p is diagonal, we have $\Sigma_p = \mathbf{U}_q^\top \Sigma_p \mathbf{U}_q = \Sigma_q$, hence $\Sigma_p = \Sigma_q$. Therefore, all equalities hold if and only if $\mu_p = \mu_q$ and $\Sigma_p = \Sigma_q$. This confirms that the proposed scoring rule is proper and strictly proper for the multivariate Gaussian distribution. $\square$

B Dataset Details

We conducted experiments on a diverse collection of real-world datasets sourced from GluonTS [44]. These datasets are commonly used for benchmarking time series forecasting models, following their default configurations in GluonTS, which include granularity, prediction horizon (Q), and the number of rolling evaluations. For each dataset, we sequentially split the data into training, validation, and testing sets, ensuring that the temporal length of the validation set matched that of the testing set. The temporal length of the testing set was based on the prediction horizon and the required number of rolling evaluations. For example, the testing horizon for the traffic dataset is calculated as

24 + 7 - 1 = 30 time steps. Consequently, the model generates 24-step predictions (Q) sequentially, with 7 distinct consecutive prediction start points, corresponding to 7 forecast instances. In our experiments, we aligned the conditioning range (P) with the prediction horizon (Q), consistent with the default setting in GluonTS (i.e., $P = Q$). Each time series was individually normalized using a scaler fitted to its own training data [15, 48]. Predictions were then rescaled to their original values for computing evaluation metrics. Table A1 summarizes the statistics of all datasets.

Table A1: Dataset summary.

Dataset	Granularity	# of time series	# of time steps	Q	Rolling evaluation
elec_au	30min	5	232,272	60	56
cif_2016	monthly	72	120	12	1
electricity	hourly	370	5,857	24	7
elec_weekly	weekly	321	156	8	3
exchange_rate	workday	8	6,101	30	5
kdd_cup	hourly	270	10,920	48	7
m1_yearly	yearly	181	169	6	1
m3_yearly	yearly	645	191	6	1
nn5_daily	daily	111	791	56	5
saugeenday	daily	1	23,741	30	5
sunspot	daily	1	73,924	30	5
tourism	quarterly	427	131	8	1
traffic	hourly	963	4,025	24	7
covid	daily	266	212	30	5
elec_hourly	hourly	321	26,304	48	7
m4_hourly	hourly	414	1,008	48	7
pedestrian	hourly	66	96,432	48	7
taxi_30min	30min	1214	1,637	24	56
uber_hourly	hourly	262	8,343	24	7
wiki	daily	2000	792	30	5

C Experiment Details

C.1 Benchmark Models

The input to benchmark models includes lagged time series values and covariates that encode time and series identification. The number of lagged values is determined by the granularity of each dataset. Specifically, we use lags of $\{1, 24, 168\}$ for hourly data, $\{1, 7, 14\}$ for daily data, and $\{1, 2, 4, 12, 24, 48\}$ for data with sub-hourly granularity. For all other datasets, only lag-1 values are used.

For datasets with hourly or finer granularity, we include the hour of the day and day of the week. For daily datasets, only the day of the week is used. Each time series is uniquely identified by a numeric identifier. All features are encoded as single values; for example, the hour of the day takes values between $[0, 23]$. These features are concatenated with the model input at each time step to form the model input vector $\mathbf{y}_t$ [4, 17].

Our method requires a state vector $\mathbf{h}_{i,t}$ to generate the parameters for the predictive distribution. To achieve this, we employ different neural architectures: RNNs and Transformer decoders, both of which maintain autoregressive properties for the multivariate autoregressive forecasting task, and MLPs for the univariate Seq2Seq forecasting task. Specifically, we use the GPVar model [4] as our RNN benchmark, the GPT model [41] for the decoder-only Transformer, and the N-HiTS model [42] for the MLPs. All models are trained to output $\mathbf{h}_{i,t}$, which is used to parameterize the predictive distribution.

C.2 Naive Baseline Description

In this paper, we use Vector Autoregression (VAR) [49] as a naive baseline model. The VAR(p) model is formulated as

$$\mathbf{z}_t = \mathbf{c} + \mathbf{A}_1 \mathbf{z}_{t-1} + \cdots + \mathbf{A}_p \mathbf{z}_{t-p} + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\epsilon), \quad (19)$$

where A_t is an $N \times N$ coefficient matrix, and c is the intercept term. In our experiments, we employ a VAR model with a lag of 1 (VAR(1)). The parameters in Eq. (19) are estimated using ordinary least squares (OLS), as described in Lütkepohl [49]. VAR models are not applied to datasets with insufficient time series in the testing set and are marked as “N/A” in this paper.

C.3 Hyperparameters

All model parameters are optimized using the Adam optimizer with L_2 regularization set to $1e^{-8}$ and gradient clipping applied at 10.0. For all methods, we cap the total number of gradient updates at 10,000 and reduce the learning rate by a factor of 2 after 500 consecutive updates without improvement. Table A2 provides the hyperparameter values that remain fixed across all datasets. In the main manuscript, we do NOT tune the hyperparameters specifically to favor the proposed loss. Instead, we use the same hyperparameters as those in GPVar [4], which were originally tuned for the log-score. Keeping the hyperparameters consistent across loss functions ensures that any observed improvements are attributable to the loss function itself rather than differences in hyperparameter settings. However, we conduct additional studies using hyperparameters tuned for each loss function in §D.4.

Table A2: Hyperparameters values.

Hyperparameter	Value
learning rate	1e-3
hidden size	40
n_layers (RNN/Transformer decoder/MLP)	2
n_heads (Transformer)	2
rank (R)	10
sampling dimension (B)	20
dropout	0.01
batch size	16

C.4 Training Procedure

Compute Resources All models were trained in an Anaconda environment using one AMD Ryzen Threadripper PRO 5955WX CPU and four NVIDIA RTX A5000 GPUs, each with 24 GB of memory.

Batch Size Following the method used in GPVar [4], we set the sample slice size to $B = 20$ time series and used a batch size of 16. Since our data sampler processes one slice of time series at a time rather than sampling 16 slices simultaneously, we set `accumulate_grad_batches` to 16, effectively achieving a batch size of 16.

Training Loop During each epoch, the model is trained on up to 400 batches from the training set, followed by the computation of the `valid_loss` on the validation set. Training is halted when one of the following conditions is met:

- A total of 10,000 gradient updates has been reached.
- No improvement in the validation set `valid_loss` is observed for 10 consecutive epochs.

The final model is the one that achieves the lowest `valid_loss` on the validation set.

Covariance Parameterization The covariance matrix Σ_t is parameterized directly by the forecasting model. Specifically, it is constructed as: $\Sigma_t = L_t L_t^T + \text{diag}(\mathbf{d}_t)$, where L_t is a low-rank matrix and $\mathbf{d}_t$ is a positive diagonal vector. This parameterization ensures that Σ_t remains positive semi-definite while being computationally efficient to learn. This parameterization is standard in probabilistic forecasting and allows the model to learn both the structure (through L_t) and scale (through $\mathbf{d}_t$) of the covariance during training. Without constraints, the MVG-CRPS loss could potentially be minimized by driving all eigenvalues of Σ_t to zero, resulting in a trivial solution. However, this is prevented through the following mechanisms:

- The diagonal entries of the covariance matrix are parameterized as $d_{i,t} = \text{softplus}(d_{i,t} + \text{diag_bias}) + \sigma_{\min}^2$, where the `softplus` function ensures that the diagonal entries are

strictly positive, regardless of the raw input values, diag_bias is initialized to approximately $\text{softplus_inv}(\sigma_{\text{init}}^2)$, ensuring that the diagonal entries are initially close to σ_{init}^2 . For instance, with $\sigma_{\text{init}} = 1.0$, the initial diagonal values start near 1.0. The addition of σ_{min}^2 provides a lower bound on the diagonal entries, ensuring that eigenvalues cannot approach zero.

The low-rank component is parameterized as $\mathbf{L}_{i,t} = \frac{\mathbf{L}_{i,t}}{\sqrt{B}}$, where dividing by rank ensures that the low-rank term is well-scaled relative to the diagonal entries. This normalization prevents the low-rank component from dominating or becoming disproportionately small in the covariance matrix.

Moreover, the MVG-CRPS loss provides a balance between the calibration and sharpness of the forecasts:

$$\mathbf{w}_t = \mathbf{S}_t^{-\frac{1}{2}} \mathbf{v}_t = \mathbf{S}_t^{-\frac{1}{2}} \mathbf{U}_t^\top (\mathbf{z}_t - \boldsymbol{\mu}_t),$$

$$\mathcal{L} = \sum_{t=1}^T \sum_{i=1}^N \sqrt{\lambda_t^i} \text{CRPS}(\Phi, w_{i,t}).$$

We observe that if the eigenvalues λ_t^i in $\mathbf{S}_t$ approach zero, $w_{i,t}$ will be scaled very aggressively. This leads to inflated residuals $w_{i,t}$, which subsequently affect the CRPS computation. Since the CRPS metric integrates over the forecast distribution $F(y)$, penalizing deviations between $F(y)$ and the empirical step function $\mathbf{1}(y > w_{i,t})$, artificially large $w_{i,t}$ values (resulting from extreme eigenvalue scaling) will cause the CRPS term to increase significantly. This behavior reflects the importance of ensuring that eigenvalues λ_t^i are well-regularized to prevent distortion in the forecast evaluation. By balancing the eigenvalue contributions, the MVG-CRPS ensures both stable calibration and sharpness in probabilistic forecasting.

SVD and Gradient Calculation We perform SVD on $\boldsymbol{\Sigma}(\mathbf{h}_t)$ to obtain $\mathbf{U}_t$ and $\mathbf{S}_t$ (the eigenvectors and eigenvalues, respectively). These are required to compute the whitening transformation: $\mathbf{w}_t = \mathbf{S}_t^{-\frac{1}{2}} \mathbf{U}_t^\top (\mathbf{z}_t - \boldsymbol{\mu}_t)$. During training, gradients of $\mathcal{L}$ need to flow back through the whitened vector $\mathbf{w}_t$, the eigenvectors matrix $\mathbf{U}_t$, the eigenvalues matrix $\mathbf{S}_t$, and the covariance matrix $\boldsymbol{\Sigma}_t$. The gradient of $\mathcal{L}$ with respect to $\mathbf{w}_t$ is $\frac{\partial \mathcal{L}}{\partial \mathbf{w}_t}$. Gradients of $\mathbf{w}_t$ are propagated to the whitening transformation: $\mathbf{w}_t = \mathbf{S}_t^{-\frac{1}{2}} \mathbf{U}_t^\top (\mathbf{z}_t - \boldsymbol{\mu}_t)$, which involves: (1) gradients with respect to $\mathbf{U}_t$; (2) gradients with respect to $\mathbf{S}_t^{-\frac{1}{2}}$ (i.e., the square root and inverse of singular values); and (3) gradients with respect to $(\mathbf{z}_t - \boldsymbol{\mu}_t)$. Using PyTorch’s `torch.linalg.svd`, we calculate the gradients of $\mathbf{U}_t$ and $\mathbf{S}_t$ via automatic differentiation. For the forward pass, the cost of SVD for $\boldsymbol{\Sigma}(\mathbf{h}_t) \in \mathbb{R}^{B \times B}$ is $O(B^3)$, where B is the matrix dimension. For the backward pass, computing the gradients of $\mathbf{U}_t$ and $\mathbf{S}_t$ also incurs $O(B^3)$ computational cost. Memory usage scales as $O(B^2)$ for storing the covariance matrix and the singular value decomposition outputs $(\mathbf{U}_t, \mathbf{S}_t)$. Additional memory is required for autograd intermediate values, scaling as $O(B^3)$. By leveraging PyTorch’s autograd system, we integrate the computation of $\mathbf{U}_t, \mathbf{S}_t$, and their gradients seamlessly into our end-to-end learning pipeline. This ensures that the whitening transformation and the loss function are fully differentiable, allowing the model parameters to be trained via gradient-based optimizers. The parameter B also plays a crucial role in the scalability of our method. By leveraging the Gaussian assumption, we are able to train the model using a much smaller subset of time series at each step. Consequently, the size of the covariance matrix is reduced to $B \times B$, as opposed to $N \times N$, where N represents the total number of time series in the dataset. This design ensures that the computational complexity of our method does not scale with N . Moreover, B is kept relatively small in our implementation (e.g., $B = 20$), making the approach computationally efficient.

C.5 Evaluation Metrics

In this paper, we repeated the evaluation procedure on the testing set ten times to compute the mean and standard deviation of each metric. For each evaluation, the metrics were calculated by averaging over all forecast instances in the testing set. For example, the reported CRPS_{sum} represents the average CRPS_{sum} across all forecast instances. Both CRPS and ES were estimated using Monte Carlo approximation based on 100 sampled predictions.

592 C.5.1 Continuous Ranked Probability Score

593 The empirical approximation of the Continuous Ranked Probability Score (CRPS) based on a finite
594 sample $\{x_1, \dots, x_n\}$ drawn from the predictive distribution F is given by:

$$\text{CRPS}(F, z) = \frac{1}{n} \sum_{i=1}^n |x_i - z| - \frac{1}{2n^2} \sum_{i=1}^n \sum_{j=1}^n |x_i - x_j|, \quad (20)$$

595 where the first term estimates the expected absolute deviation between the predictive samples and
596 the observation z , while the second term estimates the expected absolute deviation between pairs of
597 predictive samples. This Monte Carlo approximation converges to the true CRPS as $n \rightarrow \infty$. An
598 efficient empirical approximation of Eq. (20), based on a sorted sample $\{x_{(1)}, \dots, x_{(n)}\}$ from the
599 predictive distribution F , is given by:

$$\text{CRPS}(F, z) = \frac{1}{n} \sum_{i=1}^n |x_{(i)} - z| - \frac{1}{n^2} \sum_{i=1}^{n-1} i(n-i) (x_{(i+1)} - x_{(i)}), \quad (21)$$

600 where $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$ are the sorted predictive samples. The first term measures the
601 average absolute error between the sorted samples and the observation z , while the second term
602 provides a linear-time estimate of the expected pairwise absolute differences between samples,
603 avoiding the quadratic cost of a double sum. In this paper, we computed the empirical CRPS using
604 Eq. (21).

605 For a single forecast instance, we compute $\text{CRPS}_{\text{mean}}$ as the average CRPS across all time series
606 and prediction steps:

$$\text{CRPS}_{\text{mean}} = \mathbb{E}_{i,t} [\text{CRPS}(F_{i,t}, z_{i,t})], \quad (22)$$

607 where $F_{i,t}$ denotes the predictive distribution for $z_{i,t}$, represented by its empirical CDF. Since CRPS
608 evaluates one marginal distribution at a time, it does not capture joint dependencies across series. To
609 address this, we also compute CRPS_{sum} [4, 22, 23], which aggregates both forecasted and observed
610 values across all time series and applies CRPS to the resulting sums:

$$\text{CRPS}_{\text{sum}} = \mathbb{E}_t \left[\text{CRPS} \left(F_t, \sum_i z_{i,t} \right) \right], \quad (23)$$

611 where F_t is the empirical distribution formed by summing prediction samples across all time series.

612 C.5.2 Energy Score

613 The Energy Score (ES) generalizes the CRPS to evaluate distributional forecasts of vector-valued
614 random variables, making it a suitable multivariate metric for this paper:

$$\text{ES}(F, \mathbf{z}) = \frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i - \mathbf{z}\|^\beta - \frac{1}{2n^2} \sum_{i=1}^n \sum_{j=1}^n \|\mathbf{x}_i - \mathbf{x}_j\|^\beta, \quad (24)$$

615 where $\|\cdot\|$ denotes the Euclidean norm, $\mathbf{x}_i$ and $\mathbf{x}_j$ are samples from the predictive distribution, and $\mathbf{z}$
616 is the observed vector. In this paper, we set $\beta = 1$, following Ashok et al. [23]. To aggregate over the
617 prediction horizon, we compute the Frobenius norm of the forecast matrix $\|\mathbf{z}_{t+1:t+Q}\|_F$ in practice.

618 D Additional Results

619 D.1 Synthetic Data Experiment

620 We design a controlled noise experiment based on the example shown in §4.2 to evaluate the robustness
621 of different proper scoring rules when estimating parameters of a Gaussian distribution in the
622 presence of contaminated data. The experiment focuses on a two-dimensional multivariate Gaussian

623 distribution $P = \mathcal{N} \left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 & 0.8 \\ 0.8 & 4 \end{bmatrix} \right)$. From this distribution, we generate $N = 5000$ samples

624 as our base dataset. To systematically study robustness properties, we introduce contamination at
625 varying levels $\epsilon \in 0\%, 2\%, 4\%$ by randomly selecting ϵ proportion of individual data points and
626 adding a fixed offset of $+3.0$ to introduce outliers.

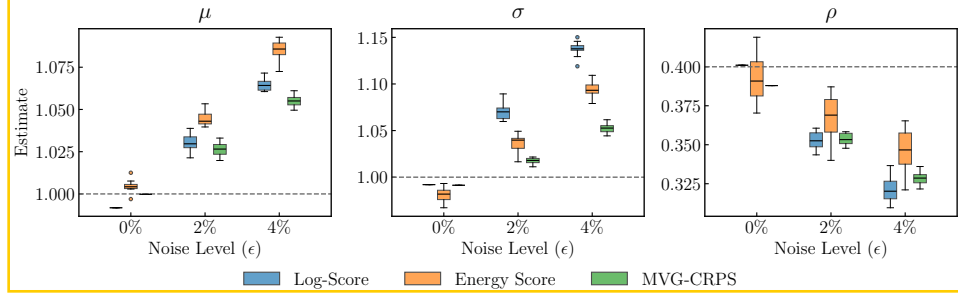


Figure A1: Parameter recovery under data contamination. Boxplots show the estimated parameters (μ, σ, ρ) of a bivariate Gaussian distribution using three proper scoring rules across varying contamination levels. Dashed lines indicate the ground truth values. Each boxplot summarizes estimates from 10 independent runs with different random seeds for contamination.

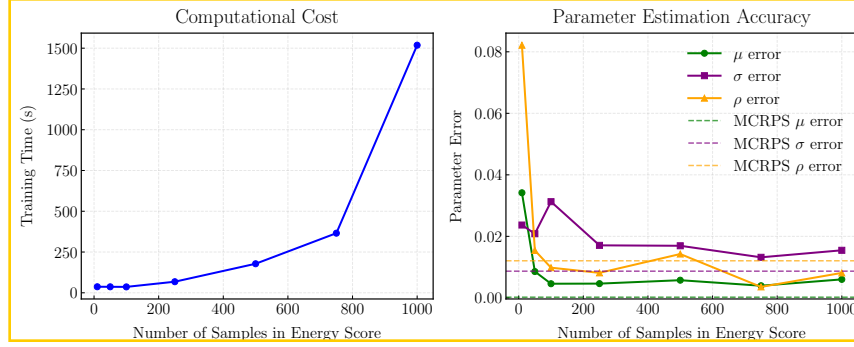


Figure A2: Computational cost versus parameter estimation accuracy for the energy score with varying sample sizes. The left panel shows training time across different numbers of Monte Carlo samples, while the right panel displays absolute errors in parameter estimates (μ, σ, ρ) , with dashed lines indicating the corresponding MVG-CRPS reference values.

This experiment compares three proper scoring rules for parameter estimation: the log-score; the energy score, implemented using a Monte Carlo approximation with 500 samples and $\beta = 1.0$; and the proposed MVG-CRPS. For each method and contamination level, we estimate three key parameters of the predictive distribution: μ (location), σ (scale), and ρ (correlation) in

$$Q = \mathcal{N} \left(\begin{bmatrix} \mu \\ -1 \end{bmatrix}, \begin{bmatrix} \sigma^2 & 2\rho\sigma \\ 2\rho\sigma & 4 \end{bmatrix} \right).$$

To ensure that parameter estimates remain within valid ranges, we apply a softplus transformation to σ and a tanh transformation to ρ , thereby constraining them to appropriate domains.

Optimization is performed using the Adam optimizer with method-specific learning rates: 3×10^{-3} for the log-score and MVG-CRPS, and 1×10^{-2} for the energy score. The number of training iterations also varies: 1000 for the log-score and MVG-CRPS, and 500 for the energy score. These hyperparameters were selected based on preliminary experiments using a validation dataset and a grid search procedure to ensure a fair comparison across methods. To assess statistical significance, we conduct 10 independent runs with different random seeds for each configuration, allowing us to examine the distribution of parameter estimates across trials.

Parameter recovery accuracy is evaluated by comparing the estimated values against the ground truth. We visualize the results using boxplots, which illustrate the distribution of estimates across runs for each method and contamination level (Fig. A1). Across all three parameters, MVG-CRPS consistently yields the most accurate and stable estimates as noise increases. For the location parameter μ and the scale σ , MVG-CRPS maintains estimates closest to the true value with minimal spread, whereas both log-score and energy score drift upward under contamination. For the correlation ρ , noise leads to downward bias for all methods, but MVG-CRPS strikes the best balance between bias and variability. The energy score appears stable under contamination, but this stability follows from its

limited sensitivity to changes in correlation, as shown in Fig. 3. Overall, MVG-CRPS shows greater robustness than the log-score and more consistent estimates than the energy score because it does not rely on Monte Carlo sampling.

Using the same example, we conducted a controlled study to examine the trade-off between computational cost and parameter estimation accuracy when using the ES with varying sample sizes. As shown in Fig. A2, training time increases monotonically with sample size due to the pairwise distance computations required by the ES. Estimation errors generally decrease with more samples but exhibit diminishing returns beyond a certain threshold (typically 100–200 samples). For reference, we include MVG-CRPS, which avoids sampling and maintains constant computational cost. Notably, even with large sample sizes (e.g., 1000), the ES does not outperform MVG-CRPS in estimation accuracy.

D.2 Other Metrics for Multivariate Autoregressive Forecasting

The results for CRPS_{mean} and ES in the multivariate autoregressive forecasting task are reported in Table A3 and Table A4, respectively. The performance of MVG-CRPS is consistent with the results reported for CRPS_{sum} in Table 1.

Table A3: Comparison of CRPS_{mean} across different scoring rules in the multivariate autoregressive forecasting task. The best scores are in boldface. MVG-CRPS scores are underlined when they are not the best overall but exceed the log-score.

VAR		GPVar			Transformer		
		log-score	energy score	MVG-CRPS	log-score	energy score	MVG-CRPS
elec_au	N/A	0.1261±0.0009	0.0887±0.0004	0.0967±0.0008	0.1633±0.0005	0.1492±0.0006	0.0793±0.0004
cif_2016	1.0000±0.0000	0.1445±0.0006	0.1690±0.0005	0.1387±0.0006	0.1611±0.0010	0.1470±0.0008	0.1178±0.0003
electricity	0.1598±0.0007	0.0601±0.0004	0.0772±0.0003	0.0623±0.0002	0.0600±0.0002	0.0705±0.0003	0.0638±0.0002
elec_weekly	0.1237±0.0009	0.1427±0.0023	0.0676±0.0008	0.0878±0.0026	0.0964±0.0022	0.0726±0.0010	0.0697±0.0012
exchange_rate	0.0070±0.0000	0.0204±0.0004	0.0094±0.0002	0.0065±0.0001	0.0112±0.0002	0.0102±0.0002	0.0115±0.0003
kdd_cup	N/A	0.3474±0.0008	0.3395±0.0011	0.2972±0.0010	0.2959±0.0008	0.4303±0.0022	0.2282±0.0005
m1_yearly	N/A	0.4397±0.0041	0.4801±0.0022	0.3566±0.0029	0.5344±0.0109	0.3291±0.0047	0.4563±0.0111
m3_yearly	N/A	0.3607±0.0084	0.2186±0.0042	0.1423±0.0053	0.3156±0.0102	0.4050±0.0061	0.2325±0.0094
nn5_daily	0.2446±0.0002	0.1525±0.0002	0.1551±0.0002	0.1540±0.0002	0.1500±0.0002	0.1453±0.0001	0.1410±0.0001
saugeenday	N/A	0.4040±0.0047	0.3733±0.0048	0.3941±0.0055	0.3771±0.0088	0.3689±0.0053	0.3705±0.0047
sunspot	N/A	18.7115±1.3296	23.3988±0.9662	17.2438±0.5833	39.7454±1.4841	16.6556±0.6167	22.6495±0.6752
tourism	0.1444±0.0007	0.2369±0.0027	0.2424±0.0010	0.2223±0.0017	0.2290±0.0010	0.2220±0.0016	0.2313±0.0017
traffic	19.9208±0.0495	0.1357±0.0002	0.1367±0.0001	0.1415±0.0001	0.1185±0.0001	0.1327±0.0001	0.1174±0.0001
Avg. Rank		2.23	2.23	1.54	2.46	1.92	1.62

Table A4: Comparison of ES across different scoring rules in the multivariate autoregressive forecasting task. The best scores are in boldface. MVG-CRPS scores are underlined when they are not the best overall but exceed the log-score.

VAR		GPVar			Transformer		
		log-score	energy score	MVG-CRPS	log-score	energy score	MVG-CRPS
elec_au ($\times 10^3$)	N/A	5.4013±0.0372	3.9136±0.0177	4.1508±0.0283	7.0039±0.0219	6.3135±0.0243	3.5217±0.0150
cif_2016 ($\times 10^3$)	125.6177±0.0000	4.2733±0.0218	4.9329±0.0161	4.1677±0.0198	4.6316±0.0270	4.1063±0.0241	3.5559±0.0145
elec ($\times 10^4$)	10.4788±0.0757	3.3124±0.0580	4.8317±0.0434	3.2435±0.0300	3.4724±0.0229	4.3757±0.0414	3.9672±0.0374
elec_weekly ($\times 10^7$)	2.2191±0.0308	2.5724±0.0799	0.8948±0.02344	1.4040±0.0887	1.5463±0.0582	0.9338±0.0308	0.9985±0.0360
exchange_rate	0.1301±0.0002	0.3972±0.0074	0.1895±0.0034	0.1216±0.0013	0.2136±0.0026	0.1774±0.0026	0.2040±0.0045
kdd_cup ($\times 10^2$)	N/A	4.7575±0.0186	4.3981±0.0164	4.0719±0.0180	4.2809±0.0134	5.9466±0.0427	3.1788±0.0122
m1_yearly ($\times 10^4$)	N/A	7.3860±0.0789	7.7576±0.0335	6.1985±0.0505	8.7079±0.1760	5.7774±0.0755	7.5130±0.1784
m3_yearly ($\times 10^3$)	N/A	3.6113±0.0703	2.2147±0.0427	1.4775±0.0495	3.1996±0.0995	4.0982±0.0621	2.4253±0.0914
nn5_daily ($\times 10^2$)	4.9419±0.0056	3.3001±0.0050	3.3004±0.0052	3.3934±0.0045	3.2546±0.0033	3.1622±0.0045	3.0996±0.0025
saugeenday ($\times 10^2$)	N/A	1.8098±0.0231	1.7135±0.0150	1.9400±0.0208	1.5780±0.0183	1.5883±0.0108	1.8043±0.0204
sunspot ($\times 10$)	N/A	2.7737±0.1195	3.1658±0.0792	2.6195±0.1003	5.4893±0.1132	2.3153±0.0467	3.2663±0.0745
tourism ($\times 10^5$)	3.5958±0.0354	6.1085±0.1132	5.6774±0.0493	5.2111±0.0896	5.0645±0.0526	4.7502±0.0585	5.2702±0.0853
traffic_nips	3358.5004±10.7535	2.2924±0.0034	2.1140±0.0023	2.2916±0.0015	2.2043±0.0012	2.2250±0.0018	2.2000±0.0018
Avg. Rank		2.46	2.00	1.54	2.38	1.92	1.69

663 D.3 Univariate Seq2Seq Forecasting

664 The results for the univariate Seq2Seq forecasting task, presented in Table A5, Table A6, and
 665 Table A7, are consistent with those from the multivariate autoregressive task. Overall, MVG-CRPS
 666 demonstrates improved accuracy compared to both the log-score and the energy score.

667 Figure A3 visualizes the output covariance matrices from models trained with different loss functions.
 668 Similar to the multivariate autoregressive task, the model trained with the log-score exhibits higher
 669 variance and covariance values, indicating greater uncertainty that may reduce forecast reliability.
 670 The figure illustrates the evolution of daily covariance in the hourly traffic dataset, shaped by both
 671 the prediction lead time and the time of day. Uncertainty tends to increase during rush hours and at
 672 longer forecast horizons. In contrast, the model trained with MVG-CRPS captures these temporal
 673 patterns while being less sensitive to extreme values, resulting in more stable estimates.

674 Figure A4 further compares probabilistic forecasts on the m4 hourly dataset. The model trained with
 675 MVG-CRPS produces narrower and better-calibrated prediction intervals than the log-score-trained
 676 model, particularly for time series with clear cyclical patterns. It also achieves higher accuracy at
 677 longer forecast horizons. These results indicate that MVG-CRPS enhances both robustness and
 678 calibration, leading to more accurate and reliable forecasts.

Table A5: Comparison of CRPS_{sum} across different scoring rules in the univariate Seq2Seq forecasting task. The best scores are in boldface. MVG-CRPS scores are underlined when they are not the best overall but exceed the log-score.

	N-HITS		
	log-score	energy score	MVG-CRPS
covid	0.1297±0.0048	N/A	0.1011±0.0022
elec_hourly	0.0470±0.0008	N/A	0.0398±0.0004
electricity	0.0409±0.0003	0.0378±0.0006	0.0372±0.0003
exchange_rate	0.0089±0.0005	0.0060±0.0002	0.0053±0.0002
m4_hourly	0.0649±0.0007	0.0595±0.0005	0.0399±0.0007
nn5_daily	0.0571±0.0003	0.0876±0.0006	0.0569±0.0004
pedestrian	0.7985±0.0511	0.9110±0.0210	0.5296±0.0071
saugenday	0.4804±0.0150	0.4372±0.0100	0.3864±0.0035
taxi_30min	0.0496±0.0002	0.0603±0.0002	0.0449±0.0001
traffic	0.2065±0.0007	0.0815±0.0001	0.0832±0.0002
uber_hourly	0.7027±0.0209	0.6461±0.0052	0.5380±0.0033
wiki	0.0660±0.0011	0.0429±0.0003	0.0465±0.0004
Avg. Rank	2.70	2.10	1.20

Table A6: Comparison of $\text{CRPS}_{\text{mean}}$ across different scoring rules in the univariate Seq2Seq forecasting task. The best scores are in boldface. MVG-CRPS scores are underlined when they are not the best overall but exceed the log-score.

	N-HITS		
	log-score	energy score	MVG-CRPS
covid	0.2076±0.0018	0.1440±0.0013	0.1022±0.0012
elec_hourly	0.0903±0.0005	0.1189±0.0004	0.0874±0.0003
electricity	0.0671±0.0002	0.0913±0.0002	0.0635±0.0001
exchange_rate	0.0173±0.0004	0.0077±0.0001	0.0073±0.0001
m4_hourly	0.1599±0.0003	0.1762±0.0007	0.1093±0.0005
nn5_daily	0.1964±0.0006	0.1588±0.0002	0.1846±0.0008
pedestrian	1.0856±0.0262	0.9254±0.0105	0.7328±0.0076
saugenday	0.4804±0.0150	0.4372±0.0100	0.3864±0.0035
taxi_30min	0.3853±0.0001	0.3939±0.0001	0.3219±0.0000
traffic	0.2514±0.0004	0.1726±0.0001	0.1583±0.0001
uber_hourly	0.9630±0.0272	0.8229±0.0062	0.6852±0.0040
wiki	0.4160±0.0006	0.2824±0.0003	0.2656±0.0002
Avg. Rank	2.67	2.25	1.08

Table A7: Comparison of ES across different scoring rules in the univariate Seq2Seq forecasting task. The best scores are in boldface. MVG-CRPS scores are underlined when they are not the best overall but exceed the log-score.

	N-HITS		
	log-score	energy score	MVG-CRPS
covid ($\times 10^5$)	2.1220 $\pm$ 0.0304	N/A	0.9401$\pm$0.0186
elec_hourly ($\times 10^5$)	0.9283 $\pm$ 0.0161	N/A	0.9088$\pm$0.0079
elec ($\times 10^5$)	0.2535 $\pm$ 0.0018	0.3123 $\pm$ 0.0019	0.2431$\pm$0.0020
exchange_rate	0.2876 $\pm$ 0.0055	0.1272 $\pm$ 0.0022	0.1240$\pm$0.0022
m4_hourly ($\times 10^4$)	0.2852 $\pm$ 0.0026	0.2890 $\pm$ 0.0029	0.2423$\pm$0.0027
nn5_daily ($\times 10^3$)	0.4170 $\pm$ 0.0018	0.3272$\pm$0.0005	0.3958 $\pm$ 0.0021
pedestrian ($\times 10^3$)	1.1571 $\pm$ 0.0177	0.9746 $\pm$ 0.0081	0.8337$\pm$0.0066
saugeenday ($\times 10^2$)	1.6690$\pm$0.0391	1.7752 $\pm$ 0.0216	1.7698 $\pm$ 0.0129
taxi_30min ($\times 10^2$)	6.9676 $\pm$ 0.0045	6.7906 $\pm$ 0.0058	5.6679$\pm$0.0004
traffic	3.6810 $\pm$ 0.0136	2.2524 $\pm$ 0.0018	2.2200$\pm$0.0022
uber_hourly	6.3252 $\pm$ 0.1785	5.4214 $\pm$ 0.0326	4.2826$\pm$0.0320
wiki ($\times 10^6$)	1.1535 $\pm$ 0.0047	0.9352 $\pm$ 0.0069	0.9338$\pm$0.0083
Avg. Rank	2.60	2.20	1.20

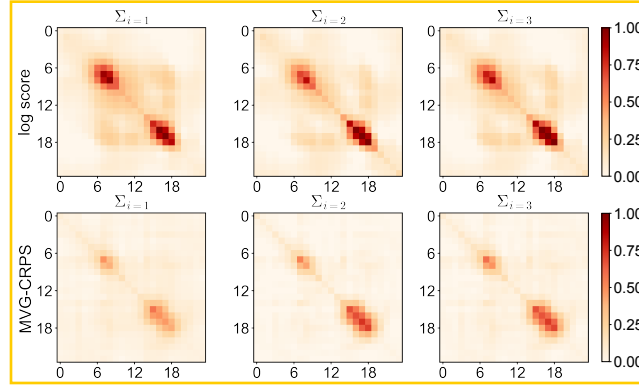


Figure A3: Comparison of output covariance matrices Σ_i from N-HITS on the traffic dataset. For visual clarity, covariance values are clipped between 0 and 1.0.

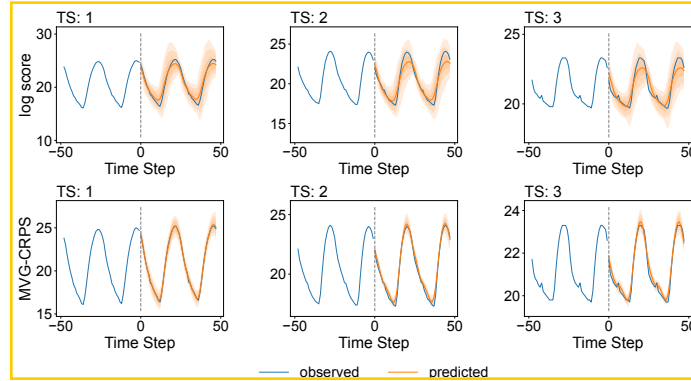


Figure A4: Comparison of probabilistic forecasts from N-HITS on the m4_hourly dataset.

679 D.4 Hyperparameter Sensitivity

680 To ensure a fair comparison, our main experiments used fixed hyperparameters across all loss
681 functions. However, since certain hyperparameters such as learning rate and rank do not affect the
682 model architecture, we performed grid searches over learning rates 10^{-4} , 10^{-3} , 10^{-2} and rank values
683 10, 20, 30 for each dataset. The optimal configuration was selected based on validation performance

for each combination of loss function, model group, and dataset. The results are presented in Table A8 and Table A9. With tuned hyperparameters, the MVG-CRPS still achieves the best average rank.

Table A8: Comparison of CRPS_{mean} across different scoring rules in the multivariate autoregressive forecasting task. The best scores are in boldface. MVG-CRPS scores are underlined when they are not the best overall but exceed the log-score. The results are obtained using models with the best hyperparameters (learning rate and rank), selected for each loss function, model group, and dataset based on validation performance. For the energy score, hyperparameter tuning was omitted due to extended training time.

VAR		GPVar			Transformer		
		log-score	energy score	MVG-CRPS	log-score	energy score	MVG-CRPS
elec_au	N/A	0.0437±0.0004	0.0887±0.0004	0.0280±0.0002	0.1158±0.0005	0.1492±0.0006	0.1410±0.0004
cif_2016	1.0000±0.0000	0.1444±0.0004	0.1690±0.0005	0.1275±0.0003	0.1217±0.0005	0.1470±0.0008	0.1201±0.0002
electricity	0.1598±0.0007	0.0601±0.0004	0.0772±0.0003	0.0665±0.0004	0.0605±0.0003	0.0705±0.0003	0.0650±0.0002
elec_weekly	0.1237±0.0009	0.1128±0.0014	0.0676±0.0008	0.1046±0.0025	0.1000±0.0020	0.0726±0.0010	0.1061±0.0013
exchange_rate	0.0070±0.0000	0.0071±0.0001	0.0094±0.0002	0.0093±0.0002	0.0131±0.0003	0.0102±0.0002	0.0161±0.0002
kdd_cup	N/A	0.3274±0.0015	0.3395±0.0011	0.2861±0.0004	0.2865±0.0012	0.4303±0.0022	0.2291±0.0010
m1_yearly	N/A	0.4883±0.0088	0.4801±0.0022	0.3333±0.0015	0.5394±0.0111	0.3291±0.0047	0.4420±0.0070
m3_yearly	N/A	0.3606±0.0133	0.2186±0.0042	0.1423±0.0053	0.3658±0.0097	0.4050±0.0061	0.2964±0.0136
nn5_daily	0.2446±0.0002	0.1474±0.0002	0.1551±0.0002	0.1510±0.0001	0.1466±0.0001	0.1453±0.0001	0.1430±0.0001
saugeenday	N/A	0.3715±0.0032	0.3733±0.0048	0.3600±0.0053	0.3756±0.0055	0.3689±0.0053	0.3831±0.0032
sunspot	N/A	10.7124±0.4618	23.3988±0.9662	16.1930±0.5734	14.4194±0.5650	16.6556±0.6167	13.1737±0.6602
tourism	0.1444±0.0007	0.2492±0.0015	0.2424±0.0010	0.1193±0.0020	0.2258±0.0020	0.2220±0.0016	0.2082±0.0014
traffic	19.9208±0.0495	0.1534±0.0002	0.1367±0.0001	0.1415±0.0001	0.1422±0.0001	0.1327±0.0001	0.1152±0.0000
Avg. Rank		2.08	2.46	1.46	2.15	2.08	1.77

Table A9: Comparison of ES across different scoring rules in the multivariate autoregressive forecasting task. The best scores are in boldface. MVG-CRPS scores are underlined when they are not the best overall but exceed the log-score. The results are obtained using models with the best hyperparameters (learning rate and rank), selected for each loss function, model group, and dataset based on validation performance. For the energy score, hyperparameter tuning was omitted due to extended training time.

VAR		GPVar			Transformer		
		log-score	energy score	MVG-CRPS	log-score	energy score	MVG-CRPS
elec_au ($\times 10^3$)	N/A	1.9601±0.0200	3.9136±0.0177	1.2546±0.0066	4.9064±0.0215	6.3135±0.0243	5.9514±0.0150
cif_2016 ($\times 10^3$)	125.6177±0.0000	4.3478±0.0127	4.9329±0.0161	3.8815±0.0072	3.6976±0.0203	4.6316±0.0270	3.5888±0.0118
elec ($\times 10^4$)	10.4788±0.0757	3.6854±0.0973	4.8317±0.0434	3.6913±0.0353	4.9963±0.0371	3.4724±0.0229	4.6774±0.0544
elec_weekly ($\times 10^7$)	2.2191±0.0308	1.9808±0.0774	0.8948±0.0234	1.2270±0.0539	1.5074±0.0600	1.5463±0.0582	1.0231±0.0402
exchange_rate	0.1301±0.0002	0.2166±0.0061	0.1895±0.0034	0.1519±0.0018	0.2317±0.0042	0.2136±0.0026	0.1569±0.0032
kdd_cup ($\times 10^2$)	N/A	4.7575±0.0186	4.3981±0.0164	5.0382±0.0142	5.2922±0.0202	4.2809±0.0134	3.2651±0.0134
m1_yearly ($\times 10^4$)	N/A	8.1941±0.1388	7.7576±0.0335	5.9567±0.0210	8.7995±0.1777	8.7079±0.1777	7.2322±0.0979
m3_yearly ($\times 10^3$)	N/A	3.6966±0.1408	2.2147±0.0427	1.4775±0.0495	3.7233±0.0925	3.1996±0.0995	2.9483±0.1275
nn5_daily ($\times 10^2$)	4.9419±0.0056	3.1966±0.0044	3.3004±0.0052	3.3303±0.0031	3.1311±0.0038	3.2546±0.0033	3.1725±0.0033
saugeenday ($\times 10^2$)	N/A	1.6529±0.0150	1.7135±0.0150	1.7678±0.0188	1.6434±0.0160	1.5780±0.0183	1.6426±0.0220
sunspot ($\times 10$)	N/A	1.7726±0.0430	3.1658±0.0792	2.5742±0.0651	2.1717±0.0468	5.4893±0.1132	1.9724±0.0363
tourism ($\times 10^5$)	3.5958±0.0354	6.5310±0.0670	5.6774±0.0493	2.8103±0.1048	6.0582±0.1162	5.0645±0.0526	4.5365±0.0662
traffic_nips	3358.5004±10.7535	2.4690±0.0026	2.1140±0.0023	2.2967±0.0012	2.2314±0.0016	2.2043±0.0012	2.1626±0.0020
Avg. Rank		2.15	2.08	1.77	2.46	2.23	1.31

D.5 Controlled Outlier Experiment

We conducted an additional experiment by injecting synthetic outliers into the training data. Specifically, a fixed proportion of observations for each sensor was perturbed with large noise ($+5\times$ the sensor's standard deviation). The test data remained clean to isolate the impact of training-time contamination. Results in Fig. A5 indicate that models trained with the log-score degrade rapidly under such noise, whereas the MVG-CRPS demonstrates greater robustness.

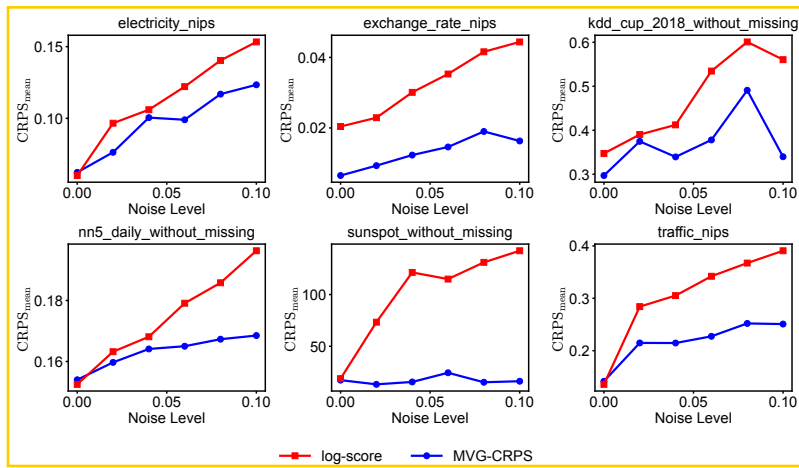


Figure A5: Controlled outlier experiment using GPVar. A fixed proportion of training samples per sensor is perturbed by adding large noise.

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